Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building and maintaining **SQL-backed applications**, debugging production issues, and delivering structured software solutions across **Java, C++, Python, JavaScript/TypeScript**, testing, and technical documentation workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed a **multi-tenant platform** on Azure SQL and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema design, indexing, and structured SQL optimization.
- Built **FastAPI** integration workflows across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation checks, debugging, and **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, handling **10K+ submissions** with modular backend services, dashboard features, and documented workflows.
- Maintained REST APIs and **AWS RDS**-backed processing pipelines with **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and maintenance flows across **3 operating systems** with **20+ unit/integration suites**.
- Analyzed and improved API and ETL logic using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **REST APIs**, WebSockets, and structured business workflows, supporting **8+ modules** and **2M+ synchronized records**.
- Modeled users, resources, invoices, and notifications in **Azure SQL** while using Jenkins-driven testing and documentation to improve release reliability and reduce wastage by **42%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **Flask**, React, TypeScript, and Firebase, supporting **50K+ booking records** with sub-**200ms** APIs, payment integrations, and audit-friendly status updates.
- Resolved workflow and webhook issues across booking and payment flows, improving booking confirmation speed by **40%** and reducing manual refund handling by **70%**.

Skills

Languages: Java, C++, Python, JavaScript/TypeScript, SQL, C#
Backend & APIs: Flask, FastAPI, Node.js, Express.js, REST APIs, Microservices
Frontend: React.js, Next.js, TailwindCSS, Java development tools
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL
Cloud & Tooling: AWS (RDS, S3), Microsoft Azure, Git, GitHub Actions, Jenkins, Docker
Engineering Practices: Structured Analysis, Technical Design, Unit & Integration Testing, Debugging, Documentation, CI/CD, Problem Solving
Foundations: Data Structures, Algorithm Design, Complexity Analysis, Team Collaboration, Learning Agility

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented **6 CPU scheduling algorithms** with configurable controls and Gantt-based output across **1K+ simulated process combinations**, strengthening algorithm design and structured debugging skills.

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, combining Java + TypeScript services, structured query-management workflows, and schema exploration from a unified interface.

HMIS Live | *MERN, TypeScript, Twilio, Appwrite*
- Built a healthcare workflow platform supporting **100+ daily appointments**, integrating structured records, monitored backend operations, and user-facing features that reduced record retrieval time by **2x**.